problem of directionality (Figure 3.14); when expressing curve trades with swaptions, we want to *exploit* the directionality (Figure 19.10).

> **Technical Point**
>
> The BPV of an option is calculated by multiplying the BPV of the underlying with the delta of the option. Thus, the BPV-neutral hedge ratio for a 2Y-10Y payer spread is given by $\frac{BPV_{10}\delta_{10}}{BPV_2\delta_2}$. If both options are ATMF (delta of 0.5), the hedge ratio of a conditional curve trade with options is the same as the hedge ratio of the underlying curve trade.

Having selected the right conditionality (1) (i.e. decided between receivers and payers), we now look at feature (2). In abstract terms, the expression of a yield curve position through options combines the yield curve with the volatility curve. In our example, the long 2Y payer versus short 10Y payer swaption position returns an up-front premium if the implied 2Y-10Y volatility curve (at entry) is upward-sloping, and requires a premium payment if it is downward-sloping.

Together, we look for conditional curve trades with options that combine the advantages of (1) and (2) (i.e. that have the right conditionality *and* provide an up-front premium payment). If we get paid a premium to execute our 2Y-10Y payer spread, we can expect to win in either directional environment: if yields increase, options are ITM, putting us into the underlying flattening trade, which should perform well in a selloff. If yields decrease, options are OTM, and we keep the initial up-front premium payment as profit (of course, we also do in the event yields increase).

As an example for a conditional butterfly trade, assume that 5Y tends to underperform versus 2Y and 10Y in the event of a selloff. Thus, we can exploit the directionality by going long 5Y payers versus short 2Y and 10Y payers and by going long 5Y receivers versus short 2Y and 10Y receivers (or by combining both by going long 5Y straddles versus short 2Y and 10Y straddles). In each case, the options are ITM only in the case of a directional move that is favorable for the underlying butterfly position. In the next step, we need to calculate whether we get paid for entering those trades (i.e. whether the curvature of the 2Y-5Y-10Y implied volatility curve is negative).

Unfortunately, as a sign of the efficiency of option markets, these opportunities are rare. The 2Y-10Y yield curve flattening in a selloff means that realized volatility in 2Y is higher than in 10Y, and if this is reflected in the implied volatility (option prices), 2Y payers will be more expensive than 10Y payers. Likewise in the butterfly example: 5Y underperforming 2Y and 10Y in a selloff is another way of saying that realized volatility in 5Y is higher than in 2Y and 10Y. Most of the time this is reflected in the option prices (i.e. in a positive curvature of the 2Y-5Y-10Y implied volatility curve). Therefore, usually strategies with the right conditionality (1) require a premium payment (2).

However, sometimes these opportunities arise and provide good candidates for trade ideas when they do. In order to screen the market for these chances, we recommend calculating for every curve steepness and butterfly trade:

(1) its directionality and the strength of that relationship (e.g. the R^2 from a regression, as in Figure 3.14);
(2) for the "right" conditionality (i.e. payers or receivers) the premium pickup or payment involved in expressing that curve trade through options.

Good candidates should have (1) a strong correlation to the direction (were the directionality to change, the conditional curve trade could lose) and (2) at least zero cost. Nhan Ngoc Le had the idea of simultaneously displaying both features of all candidates as a scatter plot chart (Figure 19.11) and looking for points in the upper-right corner. This can be a useful tool for screening the option market for trading opportunities with conditional yield curve trades. In the chart, one finds that among all possible strategies only a conditional 7Y-10Y-20Y butterfly provides stable directionality *and* a (small) premium pick-up for the right conditionality.

Similarly, one can construct conditional swap spread positions through a combination of bond options and swaptions. We have shown in Chapter 12 that before the Lehman crisis swap spreads widened (became more negative) in a selloff. Thus, if one believes that this directionality will hold true in the future, one would want to have swap spread widening positions conditional on a selloff and swap spread narrowing positions conditional on a rally, that is:

- long payer swaptions versus short bond puts;
- long receiver swaptions versus short bond calls.

Again, swap spreads widening in a selloff means that the realized volatility in swaps is greater than in bonds. Thus, the right conditionality (1) is to be always long swaptions versus short bond options. How is that realized volatility relationship reflected in the option markets? In particular, do swaptions usually trade at a premium versus bond options (2)? Do we need to pay a premium for the right conditionality? Figure 19.12 answers these questions

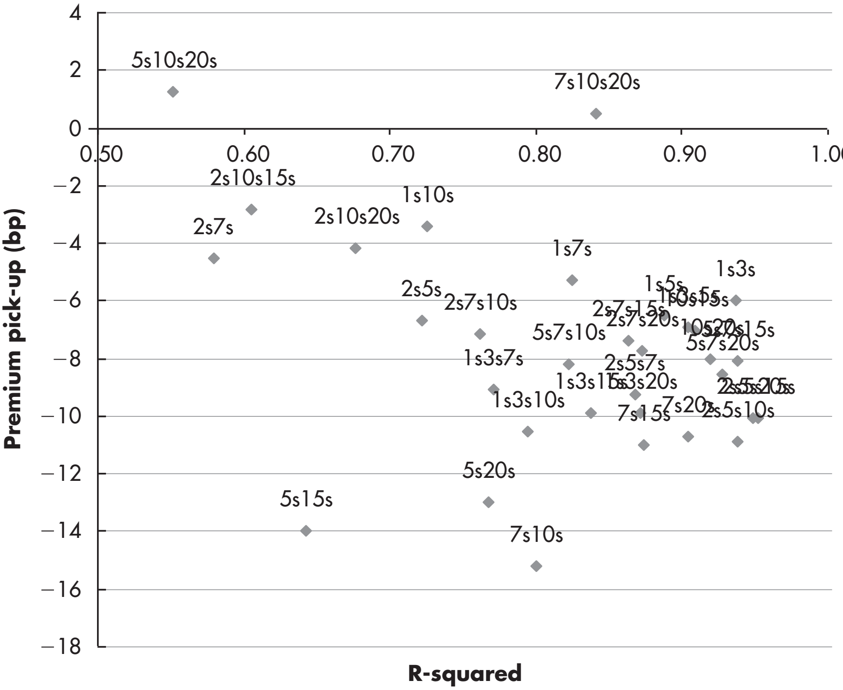

FIGURE 19.11 Premium pick-up/payment versus stability of directionality for conditional curve trades with options.
Source: Nhan Ngoc Le.

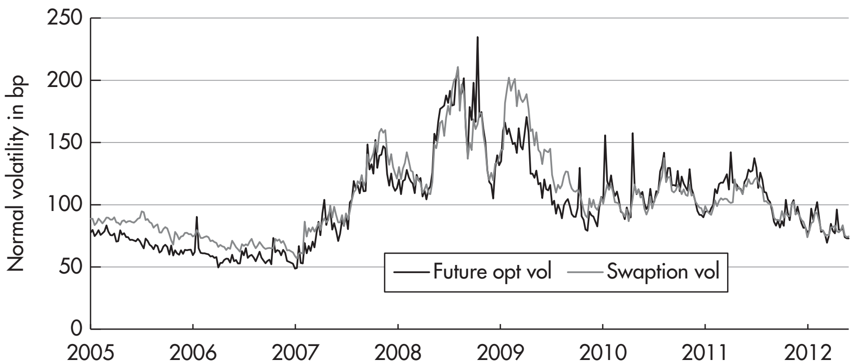

FIGURE 19.12 History of implied US bond futures option volatility versus implied USD swaption volatility (3M expiry, normalized).
Sources: data – Bloomberg; chart – Authors.

Data period: 11 May 2005 to 3 Oct 2012, weekly data.

for the US market. For reasons of liquidity and comparability, we have used futures options rather than bond options. Due to future specific flows, the volatility of implied futures options is at times higher than that of implied swaptions or implied bond options (and shows up as spikes in the chart).

It appears that the directionality of swap spreads is very well reflected in the relationship between implied swaption volatility and implied futures option volatility. Before the Lehman crisis, when swap spreads tended to widen as yields increased, swaptions normally traded at a premium versus bond options. As the Lehman crisis caused both the cost of equity for banks and thus swap spreads to widen (see Chapter 11), and the yield level to decrease, the directionality of swap spreads broke down. Correspondingly, since the Lehman crisis the premium for swaption volatility has disappeared, and it currently trades close to implied futures volatility.

An investor believing that the directionality of swap spreads will return to pre-crisis levels may see this as an opportunity to enter into long swaption versus short bond option positions.

OPTION TRADE TYPE ②: SINGLE UNDERLYING

Not delta hedging an option gives exposure to the overall realized volatility of the one period from entry to expiry, hence to the overall move of the underlying security.

Delta hedging an option (as continuous as practicable) restores the Black–Scholes framework (which assumes continuous delta hedging) outlined at the beginning of this chapter and results in the following changes versus trade type ①:

- The exposure shifts from the payoff profile of an option at expiry (the total realized volatility over the whole time period) to the continuous realized volatility of the underlying until expiry.
- The long option holder delta hedging his position is always overhedged when the market declines and underhedged when the market increases. Thus, the size of market moves (realized volatility) will determine his profit from delta hedging.
- This removes the conditionality: only the amount of moves (their frequency and scale) matters. Their direction does not. Even with a single underlying, there is no directional exposure – at least not to the direction of the underlying. (However, there is exposure to the "direction" of realized volatility.)
- The option premium (implied volatility) can be considered the market price for the right to be on the profitable side of delta hedging.

Thus, the P&L of option trade type ② is the difference between the implied volatility at entry point and the realized volatility of the underlying between entry and expiry. Put otherwise, it is the difference between the market price for the right to be on the profitable side of delta hedging and the actual profit one will realize by making use of this right. If realized volatility will exceed the current implied volatility, then the profit from delta hedging will exceed the initial option premium (i.e. the price for the right to generate profits from delta hedging).

The basic approach to the analysis of option trades of type ② is therefore to compare the two driving factors of their P&L, that is:

- the *current* implied volatility, which is a known variable;
- the *future* realized volatility (between entry and expiry), which is an unknown variable.

As the current implied volatility should reflect the market consensus about the future realized volatility, one will see value in option trades of type ② only if:

- the analyst's expectation of future realized volatility is different from the market consensus; or
- flows in the option market prevent implied volatility from reflecting the "true" market consensus about future realized volatility.

A tool to support that basic analysis could be a chart that compares the current implied volatility level with the *historical* realized volatility in different scenarios. Figure 19.13 provides an example of such a comparison for Japan.

Of course, *historical* realized volatility over the last few months can be very different from the *future* realized volatility over the next few months, which will determine the P&L. Still, the historical perspective from Figure 19.13 allows one to assess those macroeconomic environments that have in the past led to realized volatility levels that were similar to the current implied volatility. Using Figure 19.13 as an example, we conclude that the current implied volatility level was realized during 2010, when quantitative easing (expectations) removed all uncertainty (= volatility) about BoJ policy. Any shock, whether caused by nature or mankind, has resulted in a realized volatility which exceeded the current implied volatility level, at times quite significantly.

Currently, the implied volatility (19 bp) is slightly higher than the most recent realized volatility (16 bp). However, the historical perspective of

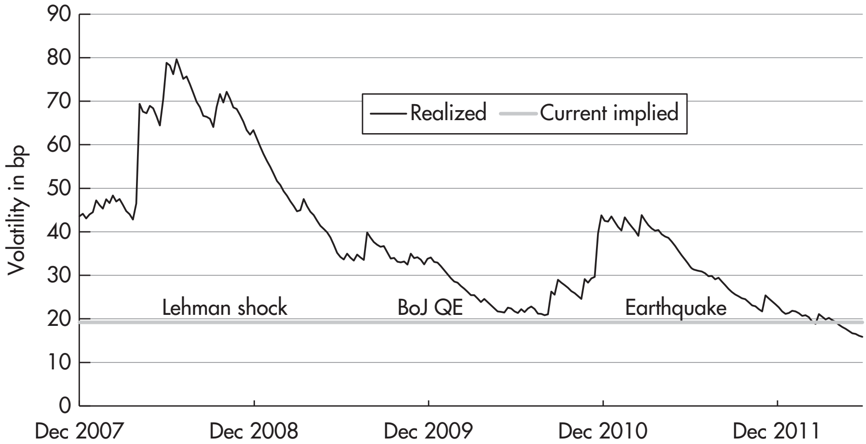

FIGURE 19.13 5Y JPY swaption volatility (normal): History of realized versus current implied volatility.
Sources: data – Bloomberg; chart – Authors.

Data period: 24 Dec 2007 to 18 Jun 2012, weekly data; "current" market data as of 18 Jun 2012.

Figure 19.13 reveals that realized volatility has usually been much higher not only than the current realized volatility but also than the current implied volatility.

Based on this tool, an analyst can now judge whether the market consensus is in line with his forecast. If he anticipates the next few months to look similar to 2010, he will find no opportunity for an option trade of type ② with a single underlying. If, on the other hand, he sees the potential for another shock, for example, a European banking crisis repeating the impact of Lehman on realized volatility, he will find value in buying options at 19 bp implied volatility. Actually, since the current level of 19 bp implied volatility seems to be close to the minimum of historical realized volatility, one could even think of long delta hedged options as a low-cost, low-risk way to position for potential shocks: if the dull period of 2010 is replicated in the next few months, delta hedging a long option should produce a profit that is about the same as the premium paid. On the other hand, any shock can result in a significantly higher realized volatility and thus profit.

Technical Points: Lognormal and Normal Volatility, Annualized Volatility, Time Horizon for Realized Volatility Calculation

In order to provide a clean analysis of realized volatility, two technical points need to be considered.

First of all, Black–Scholes assumes a (constant) lognormal distribution of yields. We do not intend to enter into a theoretical argument about the validity of this assumption[15] but instead to consider its practical pitfalls. Since lognormality assumes constant percentage changes independent of the level, the same lognormal move (percentage change) means a different absolute move (bp change) when the yield level is different. This makes comparisons of volatility on a lognormal basis meaningless in the event the yield level is different, for example, when volatilities in different parts of the yield curve (with different yield levels) are compared or the historical evolution of volatility is displayed (as the yield level typically fluctuates over time as well). Moreover, as yields approach zero, the same yield change in basis-point terms results in increasing lognormal changes.This issue is of particular relevance in the current environment of globally low interest rates.

Since lognormality cannot deal appropriately with low interest rates (and with negative yields at all), we prefer to use normal volatility. Realized normal volatility can be easily derived by calculating the standard deviation over daily basis-point changes (rather than percentage changes). This result represents the realized volatility over the time period used in the input data (e.g. trading-daily realized volatility if the input data series consists of trading-daily data). Usually, volatility is expressed in annual terms and can be obtained by multiplying the volatility for a certain time period with the square root of the number of those time units in a year. For example, trading-daily volatility can be annualized by multiplying by the square root of 252; weekly volatility can be annualized by multiplying by the square root of 52. Of course, this also works the other way round: an annual volatility can be translated into weekly and daily volatility.

If we prefer to calculate normal realized volatility, we also need to express the implied volatility in normal terms, for example, for a comparison as in Figure 19.13. However, implied volatility is usually quoted through a Black–Scholes pricing formula (i.e. in lognormal terms). The cleanest way to "normalize" it is to translate the implied lognormal volatility through the Black–Scholes formula into an option price (in dollars and cents) and then to calibrate a normal option model so that the normal

[15] Both the assumption of a *constant* lognormal volatility and of a constant *lognormal* volatility can be subjected to criticism. We deal here with the latter issue.

volatility input into the normal model replicates that option price. A faster, though less reliable method (in particular when yields are low and there is a chance of negative yields) consists in multiplying the implied lognormal volatility with the forward rate of the underlying security (with the forward horizon being given by the expiry of the option).

Second, the time window used for calculating historical realized volatility can become an issue. Commonly, analysts use a rolling window with a fixed time length (e.g. three or six months). Then, the realized volatility of a certain date is calculated as the standard deviation of the yield changes that occurred in the 3M or 6M time window prior to that date. The problem is that the result depends on the arbitrary choice of the rolling window. Imagine a time series with an average change of 1 bp per day, which contains one very volatile point with a change of 10 bp.When that volatile point enters the rolling time window, the realized volatility series spikes up. This is how it should be, since on that day there was indeed a high volatility. However, when that volatile point falls out of the rolling time window, the realized volatility series suddenly drops back to the average level. This is a problem since on the day when the sudden drop in the realized volatility occurs, nothing special has happened at all. The only thing that occurred and caused that sudden drop was that the day with an extraordinarily high 10 bp volatility happened to be precisely three or six months earlier. And depending on the arbitrary choice of the time window, the sudden drop in volatility occurs after three or after six months. This effect can be seen in Figure 19.14. What was the realized volatility in Mar 2009: 24 bp or 45 bp?

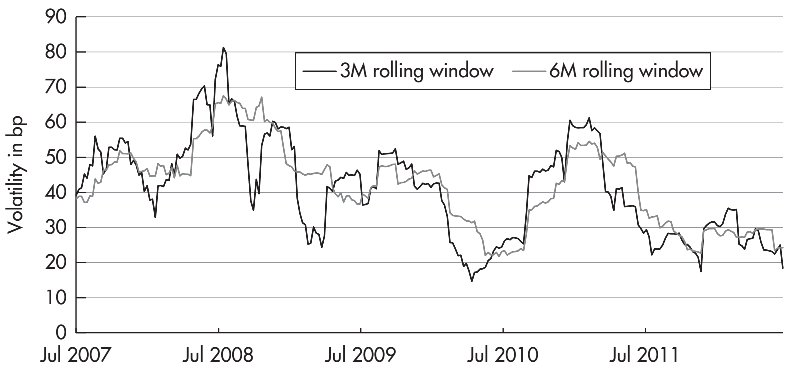

FIGURE 19.14 Realized 10Y JPY swap volatility, calculated with a 3M and a 6M rolling window.
Sources: data – Bloomberg; chart – Authors.

Data period: 2 Jul 2007 to 18 Jun 2012, weekly data.

(*continued*)

(*continued*)

When using rolling time windows, the answer depends on the arbitrary choice of 3M or 6M for the length of that window.

Consequently, using rolling time windows produces realized volatility series whose spikes higher in volatility correctly reflect market action, but whose spikes lower in volatility are the result of the arbitrary definition of the length of the time window. These can occur at different points in time and have little to do with the actual volatility on the day they occur. This problem is illustrated in Figure 19.15, which shows the realized volatility of 10Y JPY swap rates, calculated with a rolling 3M time window. The arrows indicate days on which the time series dropped sharply, not because something happened in the market but simply because the volatile day that caused the spike three months earlier rolled out of the window.

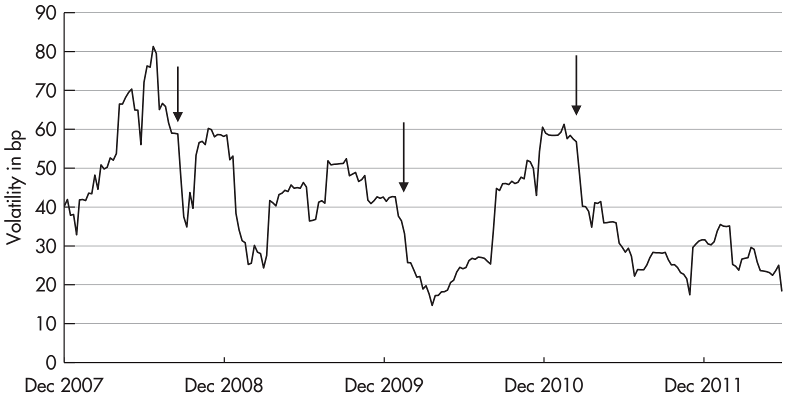

FIGURE 19.15 Realized 10Y JPY swap volatility, calculated with a 3M rolling window.
Sources: data – Bloomberg; chart – Authors.

Data period: 24 Dec 2007 to 18 Jun 2012, weekly data.

In order to mitigate this problem, we recommend replacing the fixed time window with weighting the data points with an exponentially decreasing function. For example, the latest point could have a weighting of 1, the point before that of 0.97, the point before that of 0.95, etc. The exponential weighting function assigns maximal weight to the latest data point. Thus, if this point turns out to be volatile, there will be a spike higher in the realized volatility series on that day, just as is the case with a rolling time window. On the other hand, a volatile data point will not drop out at a specific date but rather lose weight slowly and incrementally, day after day. In other words, a 3M rolling time window assigns a weighting of 1 to all points less than three months ago and of 0 to all points more than three months ago, which causes the sudden drop when a volatile point becomes more than three months old. Exponentially decreasing weightings, by contrast, assign every day a bit less importance to the volatile data point, so that its impact on the realized volatility fades away slowly. This fits the intuition of a gradually declining importance of older points for the current realized volatility quite well. Of course, there are still arbitrary decisions to be made, such as the use of an exponentially decreasing weighting function (rather than a linear or quadratic one) and the choice of a decay parameter for that function. However, these choices affect the end result much less than the sudden drops in a realized volatility series calculated with a rolling time window. To demonstrate that advantage, we have calculated the same realized 10Y JPY swap volatility history from Figure 19.14 and Figure 19.15 with exponentially decreasing weightings and depict the series in Figure 19.16. We see that upward spikes to the realized volatility series are captured as well with the exponential smoothing as they are with the rolling window. But by using exponentially decreasing weightings, we can eliminate the sharp, meaningless drops that were produced by the rolling window, replacing them with a more accurate, smoother decline.

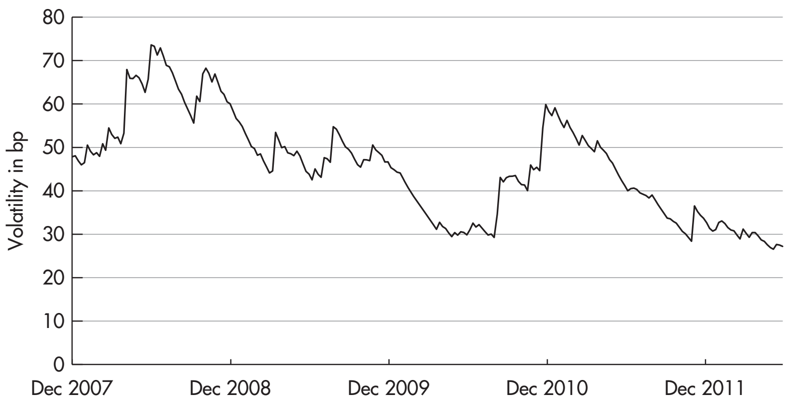

FIGURE 19.16 Realized 10Y JPY swap volatility, calculated with exponentially decreasing weightings.
Sources: data – Bloomberg; chart – Authors.

Data period: 24 Dec 2007 to 18 Jun 2012, weekly data.